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A fair solution to the compensation problem

Giacomo Valletta

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Abstract We deal with a fair division model concerning compensation among individuals endowed with different, non transferable, personal talents. We construct social orderings over all conceivable allocations based on efficiency, fairness, and robustness properties. Relying on orderings gives us the possibility of deriving the appropriate public policy even if incentive constraints arise.

1 Introduction

We study the fair allocation of a given amount of a one-dimensional, divisible resource (e.g., money) among a finite population of individuals to compensate them for their differences in a non-transferable, personal endowment. Such a personal endowment can be either an innate talent or a handicap and can be exemplified by features of human capital like health condition, bodily characteristics or social background.

For instance, one could think of the problem of a social planner willing to provide assistance to people with disabilities by distributing among them a given amount of available resources. In several countries, doctors evaluate the mental and physical condition of individuals with handicaps. An index is thus computed which determines to which extent the applicant is able to work. Some benefits are then assigned as a function of this parameter.

Recent theories of justice agree that inequalities in agents'outcomes might arise both from differences in characteristics they *should not* be held responsible for and from differences in characteristics they *should* be held responsible for. They all arrive

G. Valletta (✉)
CORE, Un. Catholique de Louvain, Louvain-la-Neuve, Belgium
e-mail: giacomo.valletta@uclouvain.be

at the common conclusion that society should only try to reduce inequalities deriving from differences of the former type.[1]

We will assume that each individual is endowed with two characteristics: her talent t, for which she is not responsible and her preferences R for which she is,[2] since they embody her subjective evaluation of her condition (namely the amount of resources she receives and her talent).

The purpose of our analysis is to look for a distribution mechanism that, on one side, allocates a given amount of available external resources M in order to correct inequalities deriving from differences in talent solely. This idea goes under the name of principle of *compensation* (Fleurbaey 1994, 1995). In performing such a compensation we cannot disregard the way individuals evaluate their condition: if all the individuals in a society deem equivalent being endowed with a given level of talent, rather than another one, then differences in talents do not need to be compensated.

On the other side, we endorse the idea that differences due to characteristics that fall into the domain of responsibility should be treated neutrally: this means that, in our case, the distribution mechanism should give the same amount of resources to individuals with the same talent regardless of the way they (subjectively) evaluate their condition. This is the so called principle of *equal resources for equal talent* (Fleurbaey 1994, 1995).

The fact that the two principles conflict with each other seems rather intuitive (Bossert 1995; Fleurbaey 1994, 1995). The latter principle prescribes a distribution of resources that does not depend on individual preferences. On the other hand preferences affect the need for a monetary compensation. Hence fully compensating agents within each preference class may lead to a distribution of resources that treats differently agents with the same talent.

To solve this problem, the literature has so far proposed allocation rules (for a complete survey of the subject see Fleurbaey and Maniquet 1999). An allocation rule specifies the set of optimal allocations as a function of the parameters of the problem: the population's profile of talents, the population's profile of preferences and the available resources. Such allocation rules can be enforced only on the basis of a full knowledge of the population's profiles of characteristics. The presence of informational constraints typically renders impossible to implement the allocation dictated by the allocation rule.

In order to overcome this difficulty we move away from this approach and we look for social ordering functions specifying a complete ranking of all the allocations as a function of the parameters of the problem. Once an ordering of the allocations is available, it is always possible to produce a precise policy recommendation, even if the social planner faces informational constraints (or any other kind of constraint).

[1] See for instance Rawls (1971), Dworkin (1981), Arneson (1989), Cohen (1989), van Parijs (1995) and Roemer (1998).

[2] Modeling responsibility by preferences it is not free of controversy. Cohen (1989) and Roemer (1998) have argued that individuals should be held responsible only for what lies within their control. According to this definition individual preferences do not necessarily belong to do domain of characteristics for which an individual should be held responsible. For an extensive discussion about the compensation-responsibility cut see Fleurbaey (2008).

This alternative approach has been already applied to several economic environments: Bossert et al. (1999) have already discussed some issues that arise if compensation schemes are to be applied in a second-best framework. In a very recent contribution Fleurbaey (2008) pushes this analysis further by providing the characterization of several social ordering functions and some insights about incentive-compatible policies. Fleurbaey and Maniquet (2006, 2007) and Luttens and Ooghe (2007) consider a model where agents have unequal production skills and different preferences and they characterize social ordering functions which satisfy properties of compensation for inequalities in skills and equal access to resources for all preferences. Maniquet and Sprumont (2004, 2005) construct orderings of allocations in economies with one private good and one partially excludable non-rival good. Fleurbaey and Maniquet (2008) construct social ordering functions in the canonical economic problem of dividing a bundle of commodities among individuals with different preferences.

We propose a way to rank alternatives based on efficiency, fairness and robustness properties. It is explained below how it is possible to embody the principle of compensation and the principle of equal resources for equal talent into a variety of specific, precise axioms. It will be clear soon that it is not possible to find a social ordering function that satisfies both principles. We show how axioms embodying the principle of compensation and axioms *weakly* embodying the principle of equal resources for equal talent together with efficiency and separability conditions (with some variants, separability conditions state that indifferent agents should not matter in the social evaluation of two alternatives) single out a specific welfare representation of agents' preferences and a specific way of ranking alternatives.

Maximizing such an ordering, under the relevant incentive constraints, gives us a specific suggestion about the way resources should be allocated. Our main finding is that when society decides, for any level of talent, what is the *fair* compensation, it should take the point of view of the individuals that are less willing to accept a low talent.

The fact that the compensation scheme we propose grants maximal priority to agents with lower talent, together with the relative scarcity of resources, also implies that it may be desirable to give a monetary compensation only to people whose level of talent is below a certain threshold.

The paper is organized as follows. Section 2 introduces the model and the relevant notation. Section 3 describes the requirements imposed on the social preferences and describes logical relations between them. Section 4 deals with social orderings. Section 5 studies the optimal allocation of resources under informational constraints. Section 6 concludes. The Appendix provides the proofs.

2 Model and main definitions

The model we analyze is derived from Fleurbaey (1994, 1995). We consider a set of economies with a finite set of agents $N \subseteq \mathbb{N}$. A given amount $M \in \mathbb{R}_{++}$ has to be shared among them through monetary transfers of amount $m_i \in \mathbb{R}_+$, for all $i \in N$. Each agent in the economy is characterized by a fixed, non transferable talent $t_i \in T$.

Let $T = [t^L, t^H] \subset \mathbb{R}$ be the set of possible talents. Moreover each agent is endowed with a personal preference ordering R_i over pairs $(m, t) \in \mathbb{R}_+ \times T$. A pair (m, t) will be called hereafter an extended bundle. This means that individuals are assumed to be able to compare extended bundles of individuals with a talent different from their own. In this framework one can think of preferences as to valuation functions: each individual is able to give a valuation of her own condition and compare it with someone else's. Even if individuals cannot actually choose between different money–talent pairs, still they are able to express a judgement about different possible conditions: one might consider the condition of other agents better or worse than her own (i.e., one might think she deserves a higher or lower compensation than what she actually receives). The individual preference ordering R_i is assumed to be continuous and strictly monotonic with respect to m_i and t_i, strict preference and indifference will be respectively denoted by P_i and I_i. Let $\mathcal{R}$ denote the set of such preferences. We will say that individual preferences $R \in \mathcal{R}$ exhibit a higher aversion to lack of talent than $R' \in \mathcal{R}$ ($R \succ_A R'$) if they satisfy the following property: for any $t, t' \in T$ and for any $m, m' \in \mathbb{R}^N_+$ we have

$$t' > t \; and (m', t') R' (m, t) \Longrightarrow (m', t') P (m, t)$$
$$t' < t \; and \; (m', t') R \; (m, t) \Longrightarrow (m', t') P' (m, t)$$

that is, any pair of indifference curves, one for R and one for R', cross at most once in the (m, t) space.

An *economy* is denoted by $e = (t_N, R_N, M)$, where $t_N = (t_i)_{i \in N}$ is the population's profile of talents, $R_N = (R_i)_{i \in N}$ is the population's profile of individual preferences and M is the available social endowment. Let us stress that the purpose of the paper is neither to establish the optimal amount M nor to establish an optimal way to tax more talented individuals in order to compensate less talented ones. This would require a more involved model that would also take into account individual productive skills and that would study the optimal tax system. This is beyond the scope of this paper. Here we do not have a market structure and the amount M is exogenously determined. Indeed in many countries a compensation is granted to disabled people independently of redistributive issues.

The domain of all economies satisfying the above assumptions will be denoted by $\mathcal{D}$. For any $e \in \mathcal{D}$, let T^e and $\mathcal{R}^e$ denote, respectively, the set of different levels of talents and the set of individual preference orderings we can find in e. An *allocation* is a vector $m_N = (m_i)_{i \in N} \in \mathbb{R}^N_+$.

The first concern of this paper is to construct a social ordering of (feasible and non feasible) allocations for all economies in the domain. A social ordering, for any given $e \in \mathcal{D}$, is a complete and transitive ranking defined over allocations. A *social ordering function* $\overline{R}$ associates every admissible economy $e = (t_N, R_N, M) \in \mathcal{D}$ with a social ordering $\overline{R}(e)$. So for an economy $e = (t_N, R_N, M) \in \mathcal{D}$ and two allocations m_N and $m'_N \in \mathbb{R}^N_+$ we write $m_N \overline{R}(e) m'_N$ to denote that m_N is (socially) at least as good as m'_N. Strict social preference and indifference will be, respectively, denoted by $\overline{P}(e)$ and $\overline{I}(e)$.

To illustrate our approach consider the simple economy depicted in Fig. 1. We only have two agents, j and k, with different talents, t_j and t_k. We consider two allocations,

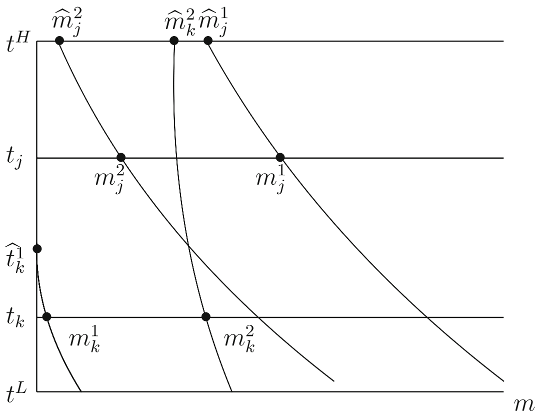


Fig. 1 The $\hat{E}$ – *maximin* function

(m_j^1, m_k^1) and (m_j^2, m_k^2) and we want to rank them. The social ordering function we propose in this paper works in the following way. For each $i \in \{j, k\}$, $q \in \{1, 2\}$ we consider the level of resource $\widehat{m}_i^q \geq 0$ which would make the agents accept the level of talent t^H instead of their current situation (m_i^q, t_i). As we can see from the figure, such a value does not exist for agent k at (m_j^1, m_k^1). In order to evaluate the situation of agent k, we then consider the lowest level of talent $\widehat{t}_k^1$ that would make her accept a null transfer instead of her current situation. More formally, $\widehat{t}_k^1$ is the level of talent such that $(0, \widehat{t}_k^1) I_k (m_k^1, t_k)$. To summarize, for each agent i we can have a continuous numerical representation of her preferences over pairs $(m, t) \in \mathbb{R}_+ \times T$:

$$\widehat{E}(m_i, t_i, R_i) = \begin{cases} \widehat{m}_i & \text{if } \widehat{m}_i \text{ exists} \\ \widehat{t}_i - t^H & \text{otherwise.} \end{cases}$$

Finally the vectors of such welfare representations are compared using the maximin rule: in our example we have that (m_j^2, m_k^2) is strictly better than (m_j^1, m_k^1) since $min\{\widehat{E}_j^2, \widehat{E}_k^2\} > min\{\widehat{E}_j^1, \widehat{E}_k^1\}$.

A social ordering function $\overline{R}$ is an $\widehat{E}$-*maximin* function if, for each economy, it prescribes an ordering of the allocations that is consistent with the application of the maximin criterion to the vectors of welfare levels generated by the function $\widehat{E}(.)$ at these allocations. More precisely, for any $e \in \mathcal{D}$ and $m_N, m'_N \in \mathbb{R}_+^N$

$$min_i \widehat{E}(m'_i, t_i, R_i) > min_i \widehat{E}(m_i, t_i, R_i) \Rightarrow m'_N \overline{P}(e) m_N.$$

3 Axioms

This section defines the properties we want to impose on our social ordering function. We start with a basic efficiency condition. The Strong Pareto axiom takes a very intuitive form here since talents are fixed and resources are one-dimensional. Notice in fact that all allocations exhausting the resources are efficient.

Strong Pareto. *For all* $e \in \mathcal{D}$ *and* $m_N, m'_N \in \mathbb{R}^N_+$ *, if* $m'_i \geq m_i$ *for all* $i \in N$*, then* $m'_N \overline{R}(e) m_N$*; if, in addition,* $m'_i > m_i$ *for some* $i \in N$*, then* $m'_N \overline{P}(e) m_N$.

The second axiom we introduce is a property of robustness of the social ordering with respect to changes in the set of agents. We require that adding or removing agents who receive the same transfer in two different allocations should not modify the social ordering. This axiom was introduced by Fleurbaey and Maniquet (1996) and it is reminiscent of the Separability condition, quite familiar in social choice, due to d'Aspremont and Gevers (1977) and of the Consistency condition, widely used in the theory of fair allocation (see Thomson 2004), except that it does not require to delete the resources consumed by the removed agents from the social endowment.

Separation. *For all* $e \in \mathcal{D}$ *and* $m_N, m'_N \in \mathbb{R}^N_+$*, if there is* $i \in N$ *such that* $m_i = m'_i$*, then*

$$m'_N \overline{R}(e) m_N \Longleftrightarrow m'_{N\setminus\{i\}} \overline{R}(t_{N\setminus\{i\}}, R_{N\setminus\{i\}}, M) m_{N\setminus\{i\}}.$$

We can now turn to the fairness conditions. We first focus on the principle of compensation. Assume there are two agents, j and k, with equal preferences and possibly a different talent. Assume also that they both consider agent j's extended bundle strictly better than agent k's. Assume finally that the monetary compensation received by agent j is reduced by a given amount and the one received by agent k is increased by the same amount but they both still consider j's extended bundle to be better than k's. The allocation we obtain must be, according to the principle of compensation, socially preferred to the former one.

Equal Preferences Transfer. *For all* $e \in \mathcal{D}$ *and* $m_N, m'_N \in \mathbb{R}^N_+$*, if there exist* $j, k \in N$ *and* $\Delta \in R_+$ *such that* $R_j = R_k$*,*

$$m'_j = m_j - \Delta,\ m'_k = m_k + \Delta \text{ and } (m'_j, t_j) P_j (m'_k, t_k),$$

with $m_i = m'_i$ *for all* $i \neq j, k$*, then* $m'_N \overline{P}(e) m_N$.

The principle implies that, ideally, two agents with equal preferences and different talents should end up on the same indifference curve (Fleurbaey 1995). Here we deal with social orderings so the axiom compares two different allocations. The aim is to reduce inequalities exclusively deriving from differences in talent.

We also consider a quite natural strengthening of this axiom: instead of requiring the preferences of the agents to be identical, we simply require that agents' indifference curves through their extended bundles in m_N and m'_N are nested. For each $i \in N$ let $L((m_i, t_i), R_i)$ and $U((m_i, t_i), R_i)$ denote respectively the (closed) lower contour set and the (closed) upper contour set of R_i at (m_i, t_i).

Nested Preferences Transfer. *For all* $e \in \mathcal{D}$ *and* $m_N, m'_N \in \mathbb{R}^N_+$ *if there exist* $j, k \in N$ *and* $\Delta \in R_+$ *such that*

$$U((m'_j, t_j), R_j) \bigcap L((m'_k, t_k), R_k) = \emptyset,$$

$m'_j = m_j - \Delta$ *and* $m'_k = m_k + \Delta$*, with* $m_i = m'_i$ *for all* $i \neq j, k$*, then* $m'_N \overline{P}(e) m_N$.

The rationale of this axiom is that, in both allocations, one of the two agents envies the other (not only at her actual talent but also at any hypothetical level of talent she may have), so performing the transfer actually reduces resources inequality.

Let us now turn to the principle of equal resources for equal talent. Assume there are two agents with the same talent but possibly different preferences who receive a different monetary compensation: performing an equalizing transfer that reduces such an inequality should be considered a social improvement.

Equal Talent Transfer. *For all $e \in \mathcal{D}$ and $m_N, m'_N \in \mathbb{R}^N_+$, if there exist $j, k \in N$ and $\Delta \in R_+$ such that $t_j = t_k$ and*

$$m_k + \Delta = m'_k < m'_j = m_j - \Delta,$$

with $m_i = m'_i$ for all $i \neq j, k$, then $m'_N \overline{P}(e) m_N$.

The principle implies that, ideally, two agents with equal talent should receive the same transfer (Fleurbaey 1995) but here we are interested in comparing two different allocations. The aim is to reduce the inequality between agents with equal talent that are treated differently.

The fairness conditions presented so far are an adaptation to this framework of the Pigou-Dalton principle of transfer, widely used in income inequality theory. Our first result shows that the requirements we have presented so far, expressing the two different ethical principles, are incompatible.

Lemma 1 *On the domain $\mathcal{D}$, no social ordering function satisfies* Equal Preferences Transfer *and* Equal Talent Transfer.

We propose now an alternative way of capturing the ethical goals of compensation and equal resources for equal talent that is inspired by Suppes' grading principles (Suppes 1966) and was introduced in the literature by Maniquet (2004). The first property, called Equal Preferences Permutation, requires that permuting the outcomes of two agents having the same preferences but possibly different talents gives us two socially equivalent allocations. Indeed, according to the principle of compensation there is no reason why society should prefer one of the two allocations: agents with the same responsibility characteristics should be treated anonymously.

Equal Preferences Permutation. *For all $e \in \mathcal{D}$ and $m_N, m'_N \in \mathbb{R}^N_+$, if there exist $j, k \in N$ such that $R_j = R_k$,*

$$(m_j, t_j) I_j (m'_k, t_k) \text{ and } (m_k, t_k) I_j (m'_j, t_j),$$

with $m_i = m'_i$ for all $i \neq j, k$, then $m_N \overline{I}(e) m'_N$.

The next axiom requires, according to the principle of equal resources for equal talent, that permuting the transfers of two agents having the same talent but possibly different preferences does not alter the value of the allocation in the social ranking. Again the justification is clear: as those agents have the same talent, they should be treated anonymously. By permuting the transfers they receive, we obtain an equally good, or equally bad, allocation.

Equal Talent Permutation. *For all $e \in D$ and $m_N, m'_N \in \mathbb{R}^N_+$, if there exist $j, k \in N$ such that $t_j = t_k$,*

$$m_k = m'_j \text{ and } m_j = m'_k,$$

with $m_i = m'_i$ for all $i \neq j, k$, then $m_N \overline{I}(e) m'_N$.

The next result shows that the incompatibility between compensation and responsibility extends beyond Lemma 1 since it has nothing to do with the degree of inequality aversion exhibited by the axioms that were involved there: also requirements like Equal Preferences Permutation and Equal Talent Permutation, which are consistent with any degree of inequality aversion (even a negative one), are in fact incompatible if combined with Strong Pareto.

Lemma 2 *On the domain $\mathcal{D}$, no social ordering function satisfies* Strong Pareto, Equal Preferences Permutation *and* Equal Talent Permutation.

Lemmas 1 and 2 are introductory results meant to show that the underlying incompatibility between the principle of compensation and the principle of equal resources for equal talent still persists when we deal with social ordering functions. The impossibility of finding a social ordering function satisfying the two principles has already been pointed out, using the same logic, by Fleurbaey and Maniquet (2005) in a model where agents have unequal skills and heterogeneous preferences over consumption and leisure.

Here we analyze a compensation minded social ordering function because we consider the principle of compensation more compelling than the principle of equal resources for equal talent. Nonetheless, as we will see later, introducing incentive-compatibility constraints considerably shapes the incompatibility between the two principles.

Interestingly, if we decide to give priority to the principle of compensation, then, combining together Nested Preferences Transfer with axioms that are essentially neutral with respect to inequality aversion, determines an extreme form of inequality aversion described by the following axiom.

Nested Preferences Priority. *For all $e \in \mathcal{D}$ and $m_N, m'_N \in \mathbb{R}^N_+$, if there exist $j, k \in N$ such that*

$$(m_j, t_j) P_j (m'_j, t_j) P_j (m'_k, t_k) P_k (m_k, t_k)$$
$$U((m'_j, t_j), R_j) \bigcap L((m'_k, t_k), R_k) = \emptyset,$$

with $m_i = m'_i$ for all $i \neq j, k$, then $m'_N \overline{P}(e) m_N$.

This property represents an extreme strengthening of Nested Preferences Transfer. It states that an inequality reduction is always socially desirable, provided that indifference curves are nested at the bundles, even when the gain from m_k to m'_k is very small and the loss from m_j to m'_j is quite large.

Lemma 3 *On the domain $\mathcal{D}$, if a SOF satisfies* Nested Preferences Transfer*,* Equal Preferences Permutation *and* Separation *then it satisfies* Nested Preferences Priority.

Results similar to this one can already be found in Fleurbaey and Maniquet (2006), Maniquet and Sprumont (2004, 2005). They all show, in different economic environments, that combining together fairness conditions that exhibit a limited aversion to inequality with robustness and efficiency conditions leads to an infinite aversion to inequality. Let us stress that we reach a similar conclusion dealing with a one-dimensional resource. This makes the contrast with the income inequality measurement framework, where combining the Pigou-Dalton transfer principle with robustness requirements does not necessarily lead to such extreme consequences (Chakravarty 1990), even stronger.

On the other hand, if we decide to give priority to the principle of equal resources for equal talent, we cannot find anything similar. In this case, in fact, we would "fall" in a purely one-dimensional framework were we are eventually free to choose the degree of inequality aversion of the social ordering function.[3]

In the last part of this section we propose two alternative weakenings of Equal Talent Transfer that represent somehow the strongest responsibility axioms that would not clash with Equal Preferences Transfer. We focus on cases where the inequality in external resources between agents with equal talent looks particularly undesirable.

The first weakening we propose takes somehow into account also individual preferences. Consider a couple of agents with the same talent who receive a different amount of resources. Moreover assume that the agent who receives less has an higher aversion to lack of talent than the other. It seems obvious that such a situation is less acceptable than the symmetric one where the "unconcerned" agent is the one who receives less. We say that it is socially desirable to perform a transfer of resources between two agents with equal talent only if the recipient's preferences exhibit a higher degree of aversion to lack of talent than the donor's ones.

Equal Talent Transfer to the Unhappy. *For all $e \in \mathcal{D}$ and $m_N, m'_N \in \mathbb{R}^N_+$, if there exist $j, k \in N$ and $\Delta \in R_+$ such that $t_j = t_k$, $R_k \succ_A R_j$ and $m_k + \Delta = m'_k < m'_j = m_j - \Delta$,* with *$m_i = m'_i$ for all $i \neq j, k$,* then $m'_N \overline{P}(e) m_N$.

The second weakening focuses on cases where one of the two agents receives a monetary compensation while the other does not. In such cases we propose a minimal redistribution requirement: it must be always possible to slightly reduce the difference in the way they are treated (i.e., it must be always possible to transfer *something* to the agent who did not receive anything in the first place) and the allocation we obtain is socially preferred to the former one.

Minimal Equal Talent Transfer. *For all $e \in \mathcal{D}$ and $m_N \in \mathbb{R}^N_+$, if there exist $j, k \in N$ such that $t_j = t_k$, $m_j = 0$, $m_k > 0$ then there exists at least one $\varepsilon \in \mathbb{R}_{++}$ and a*

[3] We could, for example, compare allocations just by computing their Gini index. This would satisfy Strong Pareto and Equal Talent Transfer (but it would violate Equal Preferences Transfer).

corresponding allocation $m'_N \in \mathbb{R}^N_+$ *such that*: $\varepsilon = m'_j < m'_k = m_k - \varepsilon, m_i = m'_i$ *for all* $i \neq j, k$, *and* $m'_N \overline{P}(e) m_N$.

Let us stress again that the two axioms are logically independent but they are both implied by Equal Talent Transfer.

4 Social orderings

In this section we show how combining together the axioms presented in the previous section determines a specific way of ranking the social alternatives.

Theorem 1 *Every social ordering function, defined on the domain* $\mathcal{D}$, *satisfying* Strong Pareto, Nested Preferences Transfer, Equal Preferences Permutation, Equal Talent Transfer to the Unhappy, Minimal Equal Talent Transfer *and* Separation *is a* $\widehat{E}$*-maximin function.*

The fact that no axiom is redundant, is shown in the appendix.

The combination of the axioms singles out a particular welfare representation of individual preferences, that is $\widehat{E}(.)$. The corresponding welfare-level vectors need then to be ranked according to the maximin criterion: a way of ranking alternatives that gives priority to agents with a low $\widehat{E}(.)$, that is, either agents with a low m_i or agents who dislike their level of talent.

The theorem does not provide a full characterization of social preferences because it does not not tell us how to rank two allocations for which $min_i \widehat{E}(m'_i, t_i, R_i) = min_i \widehat{E}(m_i, t_i, R_i)$. Nonetheless, as we will see in the next section, we can still devise a criterion to evaluate different compensation policies and, in particular, to single out the optimal one. Moreover, the refinement of the proposed social ordering obtained by applying the leximin criterion[4] to the welfare-level vectors gives us a complete ranking of the social alternatives and, as can be easily checked, satisfies all the involved axioms.

Some further discussion will help to understand how the axioms lead to the characterization result. We could in fact think of different ways to rank alternatives. For example, we might measure welfare differently keeping the same ranking criterion: take two allocations m_N and m'_N and order the extended bundles according to a reference preference $\widetilde{R} \in \mathcal{R}$ in such a way that $(m_i, t_i)\widetilde{R}(m_{i+1}, t_{i+1})$ and $(m'_i, t_i)\widetilde{R}(m'_{i+1}, t_{i+1})$ for all $i \in (1, \dots, n-1)$ then, m'_N is socially (strictly) preferred to m_N if $(m'_n, t_n)\widetilde{P}(m_n, t_n)$.[5] This social ordering function is to some extent dual to the $\widehat{E}$-*maximin* function. It fails to satisfy the compensation minded axioms (except for agents whose individual preferences are equal to the reference one) and on the contrary satisfies the responsibility minded ones (this is indeed what we would

[4] Let $\geq_L$ denote the usual leximin ordering of $\mathbb{R}^n_+$: for any $w, w' \in \mathbb{R}^n_+$, $w' \geq_L w$ if and only if the smallest coordinate of w' is greater than the smallest coordinate of w or they are equal but the second smallest coordinate of w' is greater than the second smallest coordinate of w and so on.

[5] Notice that the agents with the worst-off extended bundles with respect to $\widetilde{R}$, at the two allocations, are not necessarily the same.

have obtained if we had decided to give priority to the principle of equal resources for equal talent).

On the other hand, we might keep the welfare representation proposed in the first place but drop the maximin criterion. We could, for example, take the welfare-level vectors underlying two different allocations and calculate their Gini index. The allocation with the smallest Gini index will be socially preferred to the other one. This ranking criterion is, to some extent, compensation minded but it is not necessarily consistent with an infinite degree of inequality aversion.

Finally, the social ordering function we propose share some similarities with the *Full Health Equivalent Maximin* function proposed by Fleurbaey (2005). In a model where individuals can chose consumption and health he proposes to maximin the lowest level of full health equivalent consumption occurring in society. The full health equivalent consumption is the smallest level of consumption which an agent would accept in replacement of her actual bundle provided that she is fully healthy. Fleurbaey characterizes his criterion with four axioms: Pareto, Equal Preferences Transfer, a weaker version of Equal Talent Transfer[6] and Hansson Independence.[7]

In Fleurbaey's model individuals can freely choose the level of health. On the contrary in our model the talent is a given parameter that cannot be changed and hence it is a parameter that defines the economy. This is why, in our framework, the axioms proposed by Fleurbaey do not necessarily single out a welfare representation that is independent of the particular profile of talents in a given economy. For example, for each $i \in N$ and $m_i \in \mathbb{R}_+$ consider $\widetilde{m}_i \geq 0$ such that $(m_i, t_i) I_i (\widetilde{m}_i, t^m)$, where t^m is the maximal element of T^e. If such a value does not exist then consider the hypothetical level of talent $\widetilde{t}_i$ such that $(0, \widetilde{t}_i) I_i (m_i, t_i)$. For each $i \in N$ consider the function

$$\widetilde{E}(m_i, t_i, R_i) = \begin{cases} \widetilde{m}_i & \text{if } \widetilde{m}_i \text{ exists} \\ \widetilde{t}_i - t^m & \text{otherwise.} \end{cases}$$

Finally compare social alternatives by applying the maximin criterion to the vectors of such welfare representation of individual preferences. Such a way to rank alternatives still satisfies all the axioms proposed by Fleurbaey but the welfare representation of individual situations is profile dependent.

5 Incentive compatible allocations

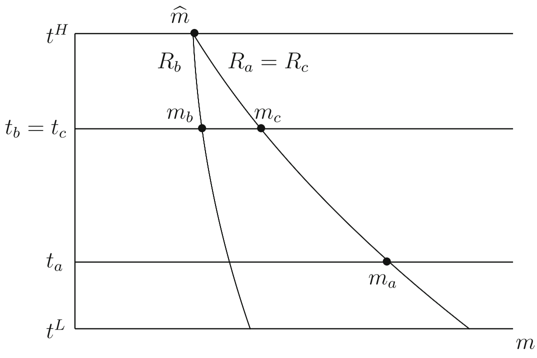
In his seminal contributions Fleurbaey (1995) characterizes two families of allocations rules that display dual properties with respect to compensation and responsibility.

$\widetilde{t}$-Egalitarian Equivalent. Choose a feasible allocation $m_N \in \mathbb{R}^+_N$ such that there exist $\widetilde{m} \in \mathbb{R}_+$, $\forall i \in N$, $(m_i, t_i) I_i (\widetilde{m}, \widetilde{t})$, or $m_i = 0$ and $(0, t_i) R_i (\widetilde{m}, \widetilde{t})$, where $\widetilde{t} \in T$ is a reference talent chosen arbitrarily.

[6] The transfer is performed only between agents with full health.

[7] The social ranking of two allocations should remain unaffected by a change in individual preferences which does not modify agents' indifference curves at the bundles they receive in these two allocations.

Fig. 2 Agent b has an incentive to misreport her preferences



$\widetilde{R}$-Conditional Equality. Choose a feasible allocation $m_N \in \mathbb{R}^{+}_{\mathbb{N}}$ such that $\forall i, j \in N$, $(m_i, t_i)\widetilde{I}(m_j, t_j)$ *or* $m_i = 0$ and $(0, t_i)\widetilde{R}(m_j, t_j)$, where $\widetilde{R} \in \mathcal{R}$ is a reference preference ordering chosen arbitrarily.

The first one is a family of compensation minded allocation rules the second one is a family of responsibility minded allocation rules.

It is easy to see that the the $\widehat{E} - maximin$ rationalizes (with some restrictions on the domain of preferences) a particular rule ($\widetilde{t} = t^H$) of the $\widetilde{t}$-Egalitarian Equivalent family of rules. At the optimum, we would obtain an allocation such that each agent is indifferent between the extended bundle she receives and a hypothetical extended bundle in which everyone receives the same transfer and everyone's talent is equal to t^H. Such an allocation would be achievable if the social planner knew the talent and the preferences of each agent. In our framework we can reasonably assume that the former is known (for example, doctors can establish the degree of disability of an individual) while individual preferences are typically not observable.[8]

Consider, for instance, the simple three-agent economy depicted in Fig. 2. At the optimum, we have full compensation between agent a and agent c. On the other hand, agent b and agent c, who have different preferences but equal talent, receive a different transfer: $m_c > m_b$. This clearly gives an incentive to agent b to misreport her preferences. We can however still use the ranking criterion we have proposed in order to choose the optimal policy available given the informational constraints.

From the previous example it is easy to understand that, if the social planner knows the distribution of individual preferences, an allocation $m_N \in \mathbb{R}^N_+$ is incentive-compatible if and only if

$$m_i = m_j \;\; for\; all\; i, j \in N \; with\; t_i = t_j.$$

Before we show the shape of the optimal incentive compatible allocation we need to introduce two more assumptions. The first one is nothing but the adaptation to this framework of the Mirrlees–Spence single crossing condition: for any couple of

[8] If the talent was not observable either, the only incentive-compatible allocation would trivially be the one that gives an equal split of the available resources to each agent.

preferences ordering R and R' in $\mathcal{R}^e$ we either have that R exhibits a higher degree of inequality aversion to lack of talent than R' or vice versa. This means that it is possible, within each economy, to rank all the preference orderings with respect to the aversion to lack of talent they exhibit.

Assumption 1 For each $R, R' \in \mathcal{R}^e$, with $R \neq R'$, either $R \succ_A R'$ or $R' \succ_A R$.

Let R^H and R^L denote the preference orderings in $\mathcal{R}^e$ with, respectively, highest and lowest aversion to lack of talent.

The second assumption we introduce is a richness requirement that looks quite reasonable specially if the number of agents in the economy is considerably large: for any class of agents with a given talent $t \in T^e$ there is at least one agent with preferences R, for each $R \in \mathcal{R}^e$.

Assumption 2 For each $(t, R) \in T^e \times \mathcal{R}^e$ there exists $i \in N$ such that $R_i = R$ and $t_i = t$.

Consider an incentive-compatible allocation, by Assumption 2, for each $t \in T^e$, there is at least one agent with such a talent and with preferences equal to R^H. It is easy to check that, among the other agents with the same talent, she is the one with lowest $\widehat{E}(.)$ (for an example see Fig. 3, for simplicity we represent an economy where there are only two kinds of individual preferences: R^L and R^H). Since we are using the maximin criterion, we should equalize, whenever possible, the lowest $\widehat{E}(.)$ values across different classes of talent.

Theorem 2 *In any economy $e \in \mathcal{D}$, under Assumptions 1 and 2, if a Social Ordering Function satisfies* Strong Pareto, Nested Preferences Transfer, Equal Preferences Permutation, Equal Talent Transfer to the Unhappy, Minimal Equal Talent Transfer *and* Separation *then, the optimal incentive compatible allocation ($m_N^{t^H}$) distributes the available resources in the following way*:

$$for\ all\ i, j \in N, \quad (m_i^{t^H}, t_i) I^H (m_j^{t^H}, t_j)\ or\ m_i^{t^H} = 0 \quad and \quad (0, t_i) R^H (m_j^{t^H}, t_j).$$

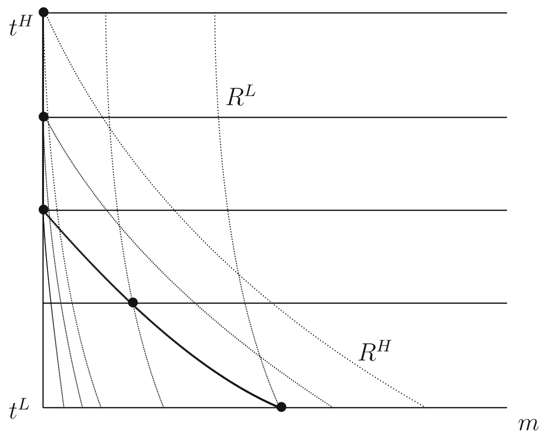


Fig. 3 Proof of Theorem 2

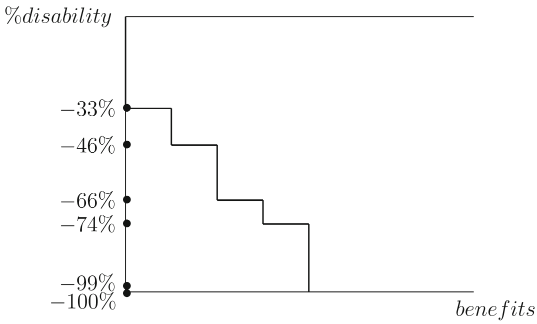


Fig. 4 The Italian social security scheme of benefits for people with disabilities

The resulting allocation is strongly affected by the incentive compatibility constraint: people with the same talent receive the same monetary transfer. The allocation we obtain by maximizing our (compensation minded) social ordering function under incentive-compatibility constraints is equivalent to a specific rule ($\widetilde{R} = R^H$) of the $\widetilde{R}$-Conditional Equality family of rules. Hence we finally obtain a first best efficient scheme where the principle of equal resources for equal talent is fully satisfied, nonetheless it shows us the highest level of priority that we can give to the principle of compensation given the informational constraints we have to face. Moreover resources are distributed taking into account the point of view of the individuals with highest aversion to lack of talent. Equality of extended resources is in fact pursued in the counterfactual situation were all agents have preferences over money-talent pairs equal to R^H.

Finally it may be impossible to fully compensate some inequalities in personal characteristics with the available resources (for example, the proportion of individuals with a low disability is typically higher than the proportion of agents with a severe disability). In that case, some inequalities persist and the better-off agents are left with no external resource, while only disadvantaged agents receive a strictly positive monetary compensation. This determines a talent threshold above which $m = 0$. Such a compensation path actually gives maximal priority to agents with a lower talent leaving a relatively big share of the population without any compensation.

This gives justification to several compensation schemes used in reality that choose to leave with no transfer people with a relatively high talent. For example, the scheme of benefits for people with disabilities used in Italy (but similar schemes are used in Belgium, France and USA) is illustrated in Fig. 4.[9] The vertical axis represents the degree of inability to work, whereas the horizontal axis represents bundles of benefits. The shape of the curve gives us a clear idea of how the compensation policy is designed: the bundle of benefits grows cumulatively as the level of handicap increases and agents with the same level of disability are equally treated. A significant part of the applicants, people with a level of disability between 0 and 32%, does not receive any kind of compensation. People with a level of disability between 33 and 73% are entitled to some monetary and non-monetary benefits that cumulate depending on the severity of their condition (i.e., specific medical supplies, priority in the unemployment, free drugs). People with a disability between 74 and 99% also receive a monthly monetary

[9] The website http://www.disabili.it gives a precise description of the compensation scheme, the figure is mine and it has only a descriptive purpose.

subsidy but only until their retirement age and provided their earnings are below a certain threshold. Individuals with a disability of 100% receive the same amount of money permanently and independently of their wealth condition.

6 Conclusions

In the first part of this paper we have examined how axioms embodying the principle of compensation and axioms weakly embodying the principle of equal resources for equal talent lead to a particular way of evaluating individual welfare and single out particular social preferences. Such social preferences grant absolute priority to agents who dislike their level of talent and to agents who receive a low compensation. The measure of individual situations obtained here (that is, $\widehat{E}(.)$) is derived from the fairness principles and the analysis did not require any other information about individual welfare than ordinal non-comparable preferences.

The second part of this paper has studied the implications of such social preferences for the evaluation of incentive-compatible policies. By maximizing a compensation minded social ordering function, under the relevant informational constraints, we anyway obtain a scheme that gives equal transfer to agents with equal talents. The fact that incentive constraints prevent the full expression of priority given to the principle of compensation does not render the principle irrelevant. On the contrary it tells us to what extent an implementable policy can take this ethical requirement into account.

Our main result is that resources should be allocated by giving maximal priority to agents with a low talent, they in fact receive the largest relative compensation one can think of.

This also implies that leaving a certain share of the population without a monetary compensation can be defended on normative grounds. In the scheme we propose such a share is endogenously determined. This result crucially depends on the particular weakening of Equal Talent Transfer we have chosen. For example, we might have opted for a different weakening entailing the arbitrary choice of a reference talent $\widetilde{t}$. Namely, to impose the full transfer condition only among agents with a talent equal to $\widetilde{t}$.

$\widetilde{t}$-Equal Talent Transfer. *For all $e \in \mathcal{D}$, $m_N, m'_N \in \mathbb{R}^N_+$, for at least one $\widetilde{t} \in T$ if there exist $j, k \in N$ and $\Delta \in R_+$ such that $t_j = t_k = \widetilde{t}$* and *$m_j - \Delta = m'_j > m'_k = m_k + \Delta$, with $m_i = m'_i$ for all $i \neq j, k$, then $m'_N \overline{P}(e) m_N$.*

If we replaced, Equal Talent Transfer to the Unhappy with $\widetilde{t}$-Equal Talent Transfer in Theorem 1 we would obtain that the welfare representation of individual utility becomes a function of the parameter $\widetilde{t}$. Namely, for each $i \in N$

$$\widehat{E}(m_i, t_i, R_i, \widetilde{t}) = \begin{cases} \widehat{m}_i & \text{if } \widehat{m}_i \text{ exists} \\ \widehat{t}_i - \widetilde{t} & \text{otherwise} \end{cases}$$

Moreover, it would be possible to adapt the Proof to theorem 1 too see that on the domain $\mathcal{D}$, for any $m_N, m'_N \in \mathbb{R}^N_+$, $m'_N \overline{P}(e) m_N$ if $min_i \widehat{E}(m'_i, t_i, R_i, \widetilde{t}) > min_i \widehat{E}(m_i, t_i, R_i, \widetilde{t})$.

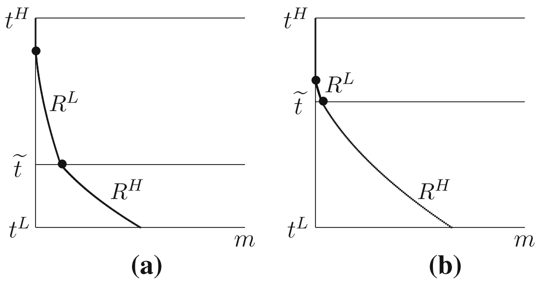


Fig. 5 A different reference talent determines a different compensation policy

The main drawback of this social ordering function is that the social ordering would depend on the (arbitrary) choice of the reference talent. This would give the social planner the freedom of choosing the level of priority to be given to agents with a severe disability. Among the agents with a talent higher than $\widetilde{t}$ the worst-off would be those with preferences equal to R^L. On the contrary, among the agents with a talent lower than $\widetilde{t}$ the worst-off would be those with preferences equal to R^H. By equalizing the lowest $\widehat{E}(.)$ values across different classes we would obtain schemes that look like those depicted in Fig. 5. Notice that the lower is the reference talent chosen by the social planner the smaller is the compensation received by people with a severe disability, the smaller is the share of population left without compensation.

Finally let us briefly discuss the assumptions underlying the analysis. First it is worth stressing that all the results obtained can be generalized to a framework without the non-negativity constraint on the monetary transfer. In particular, the first condition in the definition of $\widehat{E}(.)$ would always be satisfied. Moreover the compensation scheme we propose would eventually ask people with higher talent to pay a tax rather than receiving no transfer but still, the distribution of resources would be performed taking into account the point of view of agents with highest aversion to lack of talent. More precisely we would have that the optimal incentive compatible allocation $(m_N^{t^H})$ distributes the available resources in the following way:

$$for\ all\ i, j \in N, \quad (m_i^{t^H}, t_i) I^H (m_j^{t^H}, t_j).$$

In Fleurbaey (1995) no particular structure is assumed on the set T. Here Lemmas 1, 2 and 3 would still hold under the same assumption too (and this would be the most general domain in which they hold). On the other hand assuming that T has at least some ordinal structure plays a fundamental role in the way we built the weakening of Equal Preferences Transfer. In particular it would not be possible to talk about aversion to lack of talent without imposing a structure on T. As a consequence we obtain a welfare representation of individual preferences that also relies on such a structure.

Appendix: proofs

Proof of Lemma 1 Take the economy $e = (t, t', t', t, R', R', R, R, M) \in \mathcal{D}$, the extended bundles (m^1, t), (m^2, t), (m^3, t), (m^4, t), (m^5, t'), (m^6, t'), (m^7, t'), (m^8, t') and $\Delta \in \mathbb{R}_{++}$ with:

(1) $t \neq t'$
(2) $(m^1, t) P (m^8, t')$
(3) $(m^5, t') P' (m^4, t)$
(4) $m^4 - \Delta = m^3 > m^2 = m^1 + \Delta$
(5) $m^8 - \Delta = m^7 > m^6 = m^5 + \Delta$

We give a graphical example of such an economy in Fig. 6. By *Equal Preferences Transfer* and conditions 2, 4 and 5, $(m^1, m^8, m^5, m^3)\ \overline{P}(e)\ (m^2, m^7, m^5, m^3)$. By *Equal Talent Transfer* and condition 5, $(m^1, m^7, m^6, m^3)\ \overline{P}(e)\ (m^1, m^8, m^5, m^3)$. By *Equal Preferences Transfer* and conditions 3, 4 and 5, $(m^1, m^7, m^5, m^4)\ \overline{P}(e)\ (m^1, m^7, m^6, m^3)$. By *Equal Talent Transfer* and condition 4, $(m^2, m^7, m^5, m^3)\ \overline{P}(e)\ (m^1, m^7, m^5, m^4)$. Finally, by transitivity, $(m^1, m^7, m^6, m^3)\ \overline{P}(e)\ (m^1, m^7, m^6, m^3)$, which yields the desired contradiction. □

Proof of Lemma 2 Take the economy $e = (t, t, t', t', R, R', R', R, M) \in \mathcal{D}$, the extended bundles $(m^1, t), (m^2, t), (m^3, t), (m^4, t), (m^5, t), (m^6, t), (m^{1'}, t'), (m^{2'}, t'), (m^{3'}, t'), (m^{4'}, t'), (m^{5'}, t'), (m^{6'}, t')$ with:

(1) $t \neq t'$
(2) $m^1 > m^2 > m^3 > m^4 > m^5 > m^6$
(3) $m^{1'} > m^{2'} > m^{3'} > m^{4'} > m^{5'} > m^{6'}$
(4) $(m^1, t) I (m^{2'}, t')$ *and* $(m^6, t) I (m^{3'}, t')$
(5) $(m^2, t) I' (m^{1'}, t')$ *and* $(m^3, t) I' (m^{6'}, t')$

A graphical example of such an economy is given in Fig. 7. By *Strong Pareto* and conditions 2 and 3, $(m^1, m^4, m^{3'}, m^{5'})\ \overline{P}(e)\ (m^2, m^5, m^{4'}, m^{6'})$. By *Equal Talent Permutation* and condition 3, $(m^4, m^1, m^{3'}, m^{5'})\ \overline{I}(e)\ (m^1, m^4, m^{3'}, m^{5'})$. By *Equal Preferences Permutation* and condition 4, $(m^4, m^6, m^{2'}, m^{5'})\ \overline{I}(e)\ (m^4, m^1, m^{3'}, m^{5'})$. By *Equal Talent Permutation* and condition 3, $(m^4, m^6, m^{2'}, m^{5'})\ \overline{I}(e)\ (m^4, m^6, m^{5'}, m^{2'})$. By *Strong Pareto* and conditions 2 and 3, $(m^3, m^5, m^{4'}, m^{1'})\ \overline{P}(e)\ (m^4, m^6, m^{5'}, m^{2'})$. By *Equal Preferences Permutation* and condition 5, $(m^2, m^5, m^{4'}, m^{6'})\ \overline{I}(e)\ (m^3, m^5, m^{4'}, m^{1'})$. Finally by transitivity, we have $(m^2, m^5, m^{4'}, m^{6'})\ \overline{P}(e)\ (m^2, m^5, m^{4'}, m^{6'})$, a contradiction. □

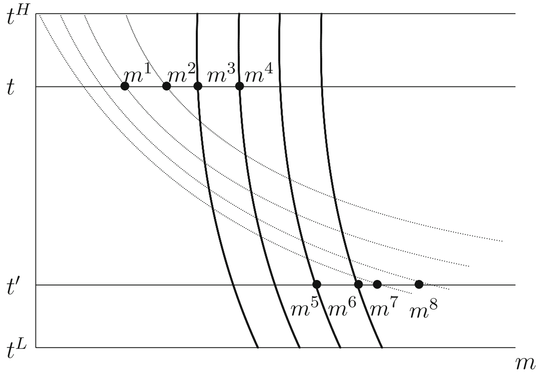


Fig. 6 Equal Preferences Transfer and Equal Talent Transfer are incompatible

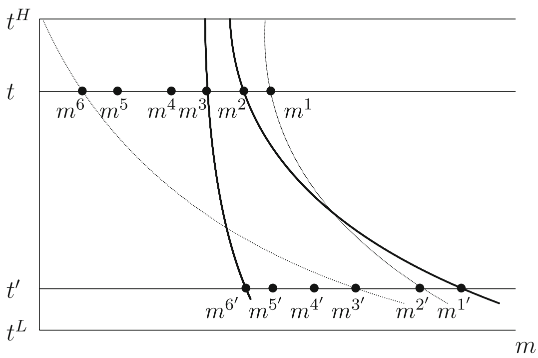


Fig. 7 Strong Pareto, Equal Preferences Permutation and Equal Talent Permutation are incompatible

Proof of Lemma 3 Let $\overline{R}$ satisfy *Nested Preferences Transfer*, *Equal Preferences Permutation* and *Separation*. Let $e = (t_N, R_N, M) \in \mathcal{D}$, $m_N, m'_N \in \mathbb{R}^N_+$, $j, k \in N$ be such that:

$$(m_j, t_j) P_j (m'_j, t_j) P_j (m'_k, t_k) P_k (m_k, t_k),$$

$$U(m'_j, R_j) \bigcap L(m'_k, R_k) = \emptyset,$$

and for all $i \neq j, k$, $m_i = m'_i$. We want to prove that $m'_N \overline{P}(e) m_N$. Let $\triangle_j = (m_j - m'_j)$ and $\triangle_k = (m'_k - m_k)$. Let $b, c \in \mathbb{N}_{++} \backslash N$, $R_b, R_c \in \mathcal{R}$, t_b, t_c, $m^1_b, m^2_b, m^1_c, m^2_c \in \mathbb{R}_+$ and $q \in \mathbb{N}_{++}$, be defined in such a way that: $R_b = R_c$; $(m^1_b, t_b) I_b (m^1_c, t_c)$ and $(m^2_b, t_b) I_b (m^2_c, t_c)$; $m^1_b = m^2_b + \frac{\triangle_k}{q}$ and $m^1_c = m^2_c + \frac{\triangle_j}{q}$; $(m^2_b, t_b) P_b (m'_k, t_k)$, $U((m^2_b, t_b), R_b) \bigcap L((m'_k, t_k), R_k) = \emptyset$, $(m'_j, t_j) P_j (m^1_b, t_b)$ and $U((m'_j, t_j), R_j) \bigcap L((m^1_b, t_b), R_b) = \emptyset$. Let $e' = (t_N, t_b, t_c, R_N, R_b, R_c, M) \in \mathcal{D}$ (an example of such a construction is given in Fig. 8).

By *Nested Preferences Transfer*,

$$\left(m_{N \backslash \{k\}}, m_k + \frac{\triangle_k}{q}, m^2_b, m^1_c \right) \overline{P}(e') \left(m_N, m^1_b, m^1_c \right).$$

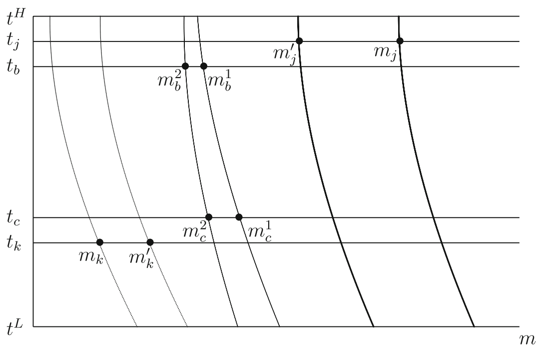


Fig. 8 From Nested Preferences Transfer to Nested Preferences Priority

By *Equal Preferences Permutation*,

$$\left(m_{N\setminus\{k\}}, m_k + \frac{\triangle_k}{q}, m_b^1, m_c^2\right) \overline{I}(e') \left(m_{N\setminus\{k\}}, m_k + \frac{\triangle_k}{q}, m_b^2, m_c^1\right).$$

By *Nested Preferences Transfer*,

$$\left(m_{N\setminus\{k,j\}}, m_k + \frac{\triangle_k}{q}, m_j - \frac{\triangle_j}{q}, \quad m_b^1, m_c^1\right) \overline{P}(e') \left(m_{N\setminus\{k\}}, \quad m_k + \frac{\triangle_k}{q}, m_b^1, m_c^2\right).$$

By transitivity,

$$\left(m_{N\setminus\{k,j\}}, \quad m_k + \frac{\triangle_k}{q}, \quad m_j - \frac{\triangle_j}{q}, m_b^1, m_c^1\right) \overline{P}(e')(m_N, m_b^1, m_c^1).$$

Replicating the argument q times yields

$$(m_{N\setminus\{k,j\}}, m_k + \triangle_k, m_j - \triangle_j, m_b^1, m_c^1)\overline{P}(e')(m_N, m_b^1, m_c^1).$$

Since $m_j - \triangle_j = m'_j$ and $m_k + \triangle_k = m'_k$, by replacing into the former equation one obtains

$$(m'_N, m_b^1, m_c^1)\overline{P}(e')(m_N, m_b^1, m_c^1).$$

Finally, by *Separation*,

$$m'_N \overline{P}(e) m_N,$$

the desired result. □

Proof of Theorem 1 Let $\overline{R}$ be a social ordering function that satisfies *Strong Pareto*, *Nested Preferences Transfer*, *Equal Preferences Permutation*, *Equal Talent Transfer to the Unhappy*, *Minimal Equal Talent Transfer* and *Separation*. By Lemma 3, the social ordering function $\overline{R}$ also satisfies *Nested Preferences Priority*. For the sake of clarity the remaining of the proof is divided in two steps.

Step 1. Consider two allocations m_N and m'_N and two different agents j and k such that for all $i \neq j, k$, $m_i = m'_i$. Let

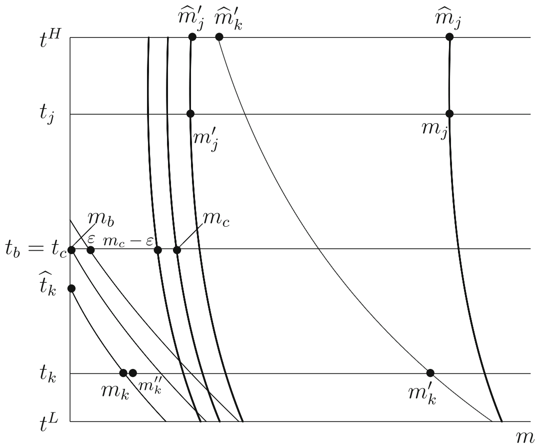
$$\widehat{E}_m = min\left\{\widehat{E}(m_j, t_j, R_j), \widehat{E}(m'_j, t_j, R_j), \widehat{E}(m'_k, t_k, R_k).\right\}$$

Without loss of generality, let us assume that $\widehat{E}(m_k, t_k, R_k) < \widehat{E}_m$. We want to prove that $m'_N \overline{P}(e) m_N$. If we also have $m'_j \geqslant m_j$ then by *Strong Pareto* we immediately get the desired result. Consider now the case $m_j > m'_j$ and assume, by contradiction, that $m_N \overline{R}(e) m'_N$. We have to examine three possible cases.

- *Case* 1: $\widehat{E}(m_k, t_k, R_k) < 0$.

  Let $b, c \in \mathbb{N}_{++}\setminus N$, $R_b, R_c \in \mathcal{R}$, $t_b, t_c \in T$, $m_b, m_c, m''_k \in \mathbb{R}_+$ be defined in such a way that: (a graphical example of the following construction is given in Fig. 9):

Fig. 9 Case 1



(1) $\widehat{t_k} < t_b = t_c < t^H$ if $\widehat{m}_m$ exists, $\widehat{t_k} < t_b = t_c < \widehat{t_m}$ otherwise
(2) $R_k = R_b$ and $R_j = R_c$
(3) $0 = m_b < m_c$ and $(m_j', t_j)P_j(m_c, t_c)$
(4) $m_k < m_k'' < m_k'$ and $(m_b, t_b)P_k(m_k'', t_k)$.

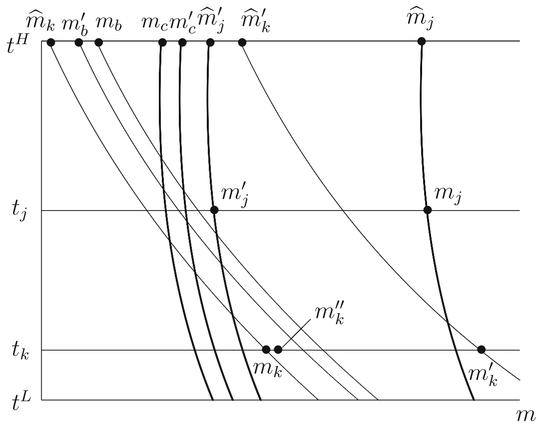
Let $e' = (t_j, t_k, R_j, R_k, M)$, $e'' = (t_j, t_k, t_b, t_c, R_j, R_k, R_b, R_c, M) \in \mathcal{D}$. Since we have assumed $m_N \overline{R}(e) m_N'$ then, by *Separation*, we have $(m_j, m_k)\ \overline{R}(e')\ (m_j', m_k')$. Take any ε such that $0 < \varepsilon < \frac{m_c}{2}$. By *Separation* again $(m_j, m_k, m_b+\varepsilon, m_c-\varepsilon)\ \overline{R}(e'')$ $(m_j', m_k', m_b+\varepsilon, m_c-\varepsilon)$. By *Nested Preferences Priority* and conditions 2 and 3, $(m_j', m_k, m_b+\varepsilon, m_c)\ \overline{P}(e'')\ (m_j, m_k, m_b+\varepsilon, m_c-\varepsilon)$. By *Nested Preferences Priority* and conditions 2 and 4, $(m_j', m_k'', m_b, m_c)\ \overline{P}(e'')\ (m_j', m_k, m_b+\varepsilon, m_c)$. By *Strong Pareto* and condition 4, $(m_j', m_k', m_b, m_c)\ \overline{P}(e'')\ (m_j', m_k'', m_b, m_c)$. By *Transitivity*, $(m_j', m_k', m_b, m_c)\ \overline{P}(e'')\ (m_j', m_k', m_b+\varepsilon, m_c-\varepsilon)$. This is true for each ε such that $0 < \varepsilon < \frac{m_c}{2}$ hence, given condition 1 and 3, it represents a violation of *Minimal Equal talent Transfer* and yields the desired contradiction so that $m_N' \overline{P}(e) m_N$

- *Case* 2: $\widehat{E}(m_k, t_k, R_k) \geq 0$ and $R_k \succ_A R_j$.
  Let $b, c \in \mathbb{N}_{++} \setminus N$, $R_b, R_c \in \mathcal{R}$, $t_b, t_c \in T$, $m_b, m_b', m_c, m_c', m_k'' \in \mathbb{R}_+$, $\Delta \in \mathbb{R}_{++}$ be defined in such a way that (see Fig. 10):

(1) $t_b = t_c = t^H$
(2) $R_k = R_b$ and $R_j = R_c$
(3) $\widehat{E}(m_k, t_k, R_k) < m_b' < m_c' < \widehat{E}_m$ and $m_b' + \Delta = m_b \leq m_c = m_c' - \Delta$
(4) $m_k < m_k''$ and $(m_b, t_b)P_k(m_k'', t_k)$.

Let $e' = (t_j, t_k, R_j, R_k, M)$, $e'' = (t_j, t_k, t_b, t_c, R_j, R_k, R_b, R_c, M) \in \mathcal{D}$. As in the previous proof we assume, $m_N \overline{R}(e) m_N'$. By applying *Separation* twice we have $(m_j, m_k)\ \overline{R}(e')\ (m_j', m_k')$ and $(m_j, m_k, m_b, m_c)\ \overline{R}(e'')\ (m_j', m_k', m_b, m_c)$. By *Nested Preferences Priority* and conditions 2, 3 and 4, $(m_j', m_k, m_b, m_c')\ \overline{P}(e'')\ (m_j, m_k,$

Fig. 10 Case 2



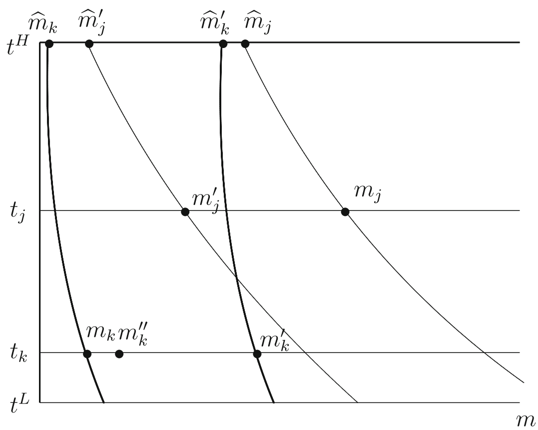
m_b, m_c) and $(m'_j, m''_k, m'_b, m'_c)\ \overline{P}(e'')\ (m'_j, m_k, m_b, m'_c)$. By *Strong Pareto* and condition 4, $(m'_j, m'_k, m'_b, m'_c)\ \overline{P}(e'')\ (m'_j, m''_k, m'_b, m'_c)$. By *Transitivity*, (m'_j, m'_k, m'_b, m'_c) $\overline{P}(e'')\ (m'_j, m'_k, m_b, m_c)$, which, given conditions 1 and 2, is a violation of *Equal Talent Transfer to the Unhappy* since, by construction, $R_b \succ_A R_c$. This yields the desired contradiction so that $m'_N \overline{P}(e) m_N$.

- *Case* 3: $\widehat{E}(m_k, t_k, R_k) \geq 0$. and $R_j \succ_A R_k$.

If we have $\widehat{E}(m'_k, t_k, R_k) < \widehat{E}(m'_j, t_j, R_j)$ then directly by *Nested Preference Transfer* we get the desired result (Fig. 11).

On the other hand, assume $\widehat{E}(m'_k, t_k, R_k) \geq \widehat{E}(m'_j, t_j, R_j)$ and $m_N \overline{R}(e) m'_N$. Consider m''_k such that $m_k < m''_k < m'_k$, $(m'_j, t_j) P_j\ (m''_k, t_k)$ and $U((m'_j, t_j), R_j) \bigcap L((m''_k, t_k), R_k) = \emptyset$. Let $e' = (t_j, t_k, R_j, R_k, M)$. By *Separation* we have $(m_j, m_k)\ \overline{R}(e')$ (m'_j, m'_k). By *Nested Preferences Priority*, $(m''_k, m'_j)\overline{P}(e')\ (m_k, m_j)$. By *Strong Pareto*, $(m'_k, m'_j)\ P(e')(m_k, m_j)$ which yields the desired contradiction.

Fig. 11 Case 3



***Step* 2**. Take now two allocations m_N and m'_N such that

$$min_i \widehat{E}(m'_i, t_i, R_i) > min_i \widehat{E}(m_i, t_i, R_i).$$

Then, by monotonicity of preferences, one can find two allocations $\overline{m}_N, \overline{m}'_N$ such that for all i, $m_i < \overline{m}_i, \overline{m}'_i < m'_i$, and there exists i_0 such that for all $i \neq i_0$

$$\widehat{E}(\overline{m}_i, t_i, R_i) > \widehat{E}(\overline{m}'_i, t_i, R_i) > \widehat{E}(\overline{m}'_{i_0}, t_{i_0}, R_{i_0}) > \widehat{E}(\overline{m}_{i_0}, t_{i_0}, R_{i_0}).$$

Let $Q = N \backslash \{i_0\}$ and consider a sequence of allocations $(\overline{m}^q_N)_{1 \leq q \leq |Q|+1}$ such that

$$\begin{aligned} \overline{m}^q_i &= \overline{m}'_i \quad \text{for all } i \in Q \text{ such that } i < q \\ \overline{m}^q_i &= \overline{m}_i \quad \text{for all } i \in Q \text{ such that } i \geq q \end{aligned}$$

while

$$\overline{m}'_{i_0} = \overline{m}^{|Q|+1}_{i_0} > \overline{m}^{|Q|}_{i_0} > \cdots > \overline{m}^1_{i_0} = \overline{m}_{i_0}.$$

This entails that for all $q \in Q$

$$\widehat{E}(\overline{m}^q_q, t_m, R_m) > \widehat{E}(\overline{m}^{q+1}_q, t_m, R_k) > \widehat{E}(\overline{m}^{q+1}_{i_0}, t_{i_0}, R_{i_0}) > \widehat{E}(\overline{m}^q_{i_0}, t_{i_0}, R_{i_0}),$$

while for all $q \in Q$, and all $i \neq i_0, q$, $\overline{m}^q_i = \overline{m}^{q+1}_i$. By Step 1, $\overline{m}^{q+1}_N P(e) \overline{m}^q_N$ for all $q \in Q$. By *Strong Pareto*, $m'_N P(e) \overline{m}^{|Q|+1}_N$ and $\overline{m}^1_N P(e) m_N$. By transitivity, $m'_N P(e) m_N$. □

Examples of social ordering functions satisfying all the axioms but:

(1) *Strong Pareto*. Compare the welfare-level vectors using the lexicographic minimax criterion: a distribution is at least as good as another one if its maximum is lower, or they are equal and the second highest value is lower, or... or they are all equal.

(2) *Nested Preferences Transfer*. Let $\overline{R}$ be defined by: for all $e \in \mathcal{D}, m_N, m'_N \in \mathbb{R}^+_N$, if

$$(\widehat{E}(m'_i, t_i, R_i))_{i \in N} >_L (\widehat{E}(m_i, t_i, R_i))_{i \in N}$$

and there exist $k, k' \in N$ with $t_k \neq t_{k'}$ such that

$$\begin{aligned} 0 &< \widehat{E}(m_k, t_k, R_k) < \widehat{E}(m'_k, t_k, R_k) < \widehat{E}(m'_{k'}, t_{k'}, R_{k'}) \\ &< \widehat{E}(m_{k'}, t_{k'}, R_{k'}), \end{aligned}$$

with $\widehat{E}(m'_i, t_i, R_i) = \widehat{E}(m_i, t_i, R_i)$ for all $i \neq k, k'$, then $m'_N \overline{I}(e) m_N$. In all other cases $m'_N \overline{P}(e) m_N$. If

$$(\widehat{E}(m'_i, t_i, R_i))_{i \in N} =_L (\widehat{E}(m_i, t_i, R_i))_{i \in N}$$

then $m'_N \overline{I}(e) m_N$.

(3) *Equal Preferences Permutation*. Let $\overline{R}$ be defined by: for all $e \in \mathcal{D}$, $m_N, m'_N \in \mathbb{R}^+_N$, if

$$(\widehat{E}(m'_i, t_i, R_i))_{i \in N} >_L (\widehat{E}(m_i, t_i, R_i))_{i \in N}$$

or

$$(\widehat{E}(m'_i, t_i, R_i))_{i \in N} =_L (\widehat{E}(m_i, t_i, R_i))_{i \in N}$$

and there exist $k, k' \in N$ with $t_{k'} \neq t_k$ such that $\widehat{E}(m_k, t_k, R_k) < \widehat{E}(m_i, t_i, R_i)$ for all $i \neq k$; $\widehat{E}(m'_{k'}, t_{k'}, R_{k'}) < \widehat{E}(m'_i, t_i, R_i)$ for all $i \neq k'$, then $m'_N \overline{P}(e) m_N$. In all other cases $m'_N \overline{I}(e) m_N$.

(4) *Equal Talent Transfer to the Unhappy and Minimal Equal Talent Transfer.* Let $\overline{R}$ be defined by: for all $e \in \mathcal{D}$, $m_N, m'_N \in \mathbb{R}^+_N$, if

$$(\widehat{E}(m'_i, t_i, R_i))_{i \in N} >_L (\widehat{E}(m_i, t_i, R_i))_{i \in N}$$

and there exist $k, k' \in N$ with $R_{k'} \neq R_k$ such that

$$\widehat{E}(m_k, t_k, R_k) < \widehat{E}(m'_k, t_k, R_k) < \widehat{E}(m'_{k'}, t_{k'}, R_{k'}) < \\ \widehat{E}(m_{k'}, t_{k'}, R_{k'}),$$

with $\widehat{E}(m'_i, t_i, R_i) = \widehat{E}(m_i, t_i, R_i,)$ for all $i \neq k, k'$ and $U((m'_{k'}, t_{k'}), R_{k'}) \bigcap L((m'_k, t_k), R_k) \neq \emptyset$, then $m'_N \overline{I}(e) m_N$. In all other cases $m'_N \overline{P}(e) m_N$. If

$$(\widehat{E}(m'_i, t_i, R_i))_{i \in N} =_L (\widehat{E}(m'_i, t_i, R_i))_{i \in N}$$

then $m'_N \overline{I}(e) m_N$.

(5) *Separation*. See example page 11 paragraph 4.

Proof of Theorem 2 By Theorem 1 we know that if a Social Ordering Function satisfies the listed axioms then, for any $m'_N, m_N \in \mathbb{R}^N_+$,

$$min_i \widehat{E}(m'_i, t_i, R_i) > min_i \widehat{E}(m_i, t_i, R_i) \Longrightarrow m'_N \overline{P}(e) m_N$$

At $m^{t^H}_N$, given Assumption 1 and 2, it holds true that the worst-off agents are (for a graphical example see Fig. 3) all $k \in N$ such that $R_k = R^H$ and $m_k > 0$:

$$min_i \{\widehat{E}(m^{t^H}_i, t_i, R_i)\}_{i \in N} = \widehat{E}(m^{t^H}_k, t_k, R^H).$$

Assume, by contradiction, that there exists an incentive compatible allocation m'_N such that $min_i \widehat{E}(m'_i, t_i, R_i) > min_i \widehat{E}(m^{t^H}_i, t_i, R_i)$. This implies that, for all $j \in N$ such that $R_j = R^H$ and $m'_j > 0$, $m'_j > m^{t^H}_j$. By incentive compatibility we should then increase by the same amount the transfer received by all the other agents having the same level of talent but different preferences. This contradicts that, by construction, $m_N^{t^H}$ is an allocation exhausting the available resources. $\square$

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